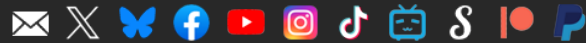


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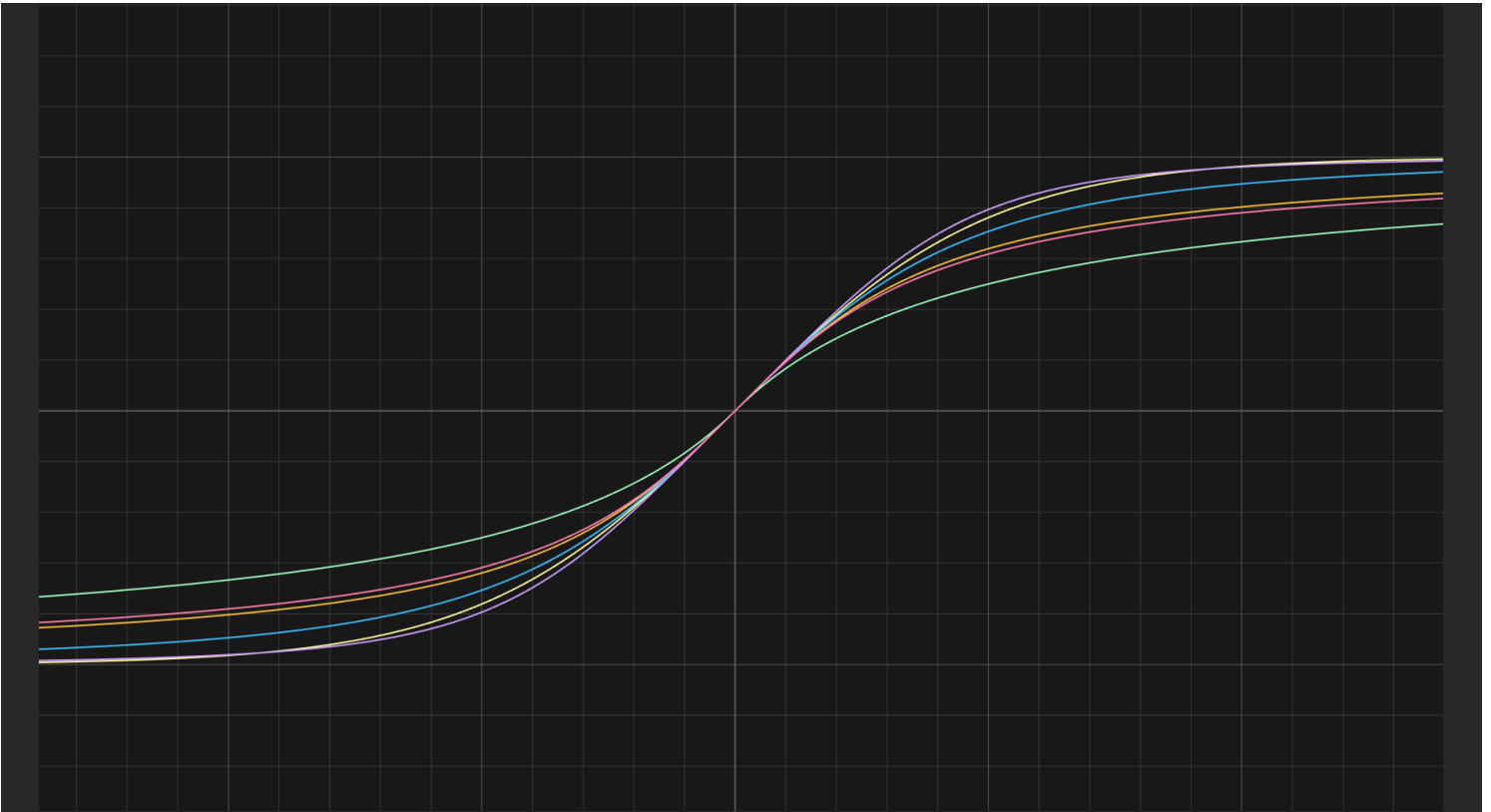


## Intro

In this article I collected a list of sigmoid functions. All of them can/should be stretched to achieve any desired inclination or slope at the origin, simply by multiplying  $x$  by the necessary constant.

| Type          | Sigmoid                                             | Inverse                                          | Derivative                                      |
|---------------|-----------------------------------------------------|--------------------------------------------------|-------------------------------------------------|
| Trigonometric | $\frac{2}{\pi} \arctan\left(\frac{\pi}{2} x\right)$ | $\frac{2}{\pi} \tan\left(\frac{\pi}{2} x\right)$ | $\frac{4}{4 + \pi^2 x^2}$                       |
| Hyperbolic    | $\tanh(x) = 1 - \frac{2}{1 + e^{2x}}$               | $\frac{1}{2} \ln \frac{1+x}{1-x}$                | $\frac{4}{(e^x + e^{-x})^2}$                    |
| S1            | $\frac{x}{1 +  x }$                                 | $\frac{x}{1 -  x }$                              | $(1 +  x )^{-2}$                                |
| S2            | $\frac{x}{\sqrt{1 + x^2}}$                          | $\frac{x}{\sqrt{1 - x^2}}$                       | $(1 + x^2)^{-3/2}$                              |
| S3            | $\frac{x}{\sqrt[3]{1 + x^2  x }}$                   | $\frac{x}{\sqrt[3]{1 - x^2  x }}$                | $(1 + x^2  x )^{-4/3}$                          |
| T1            | $\frac{x}{1 + \sqrt{1 + x^2}}$                      | $\frac{2x}{1 - x^2}$                             | $\frac{1}{\sqrt{1 + x^2} (1 + \sqrt{1 + x^2})}$ |

Below goes a visual comparison of the different sigmoids together, together with a [Graphtoy link](#).



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